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Secure Trade Finance Management system

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Secure Trade Finance Management system

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Abstract— Secure Trade Finance Management System is a web-based application developed to manage trade finance documents, transactions, users, risk scores, alerts, ledgers, and reports. The system uses a React frontend and FastAPI backend with PostgreSQL database support, Redis caching, JWT authentication, and role-based access control. Trade documents are uploaded, stored, hashed using SHA-256, and verified through document integrity checks. Transactions are created between buyers and sellers and move through statuses such as Pending, In Progress, Completed, Disputed, and Declined. The system also includes risk score calculation using internal transaction signals, alerts, country and industry inputs, and external risk signal connectors. Dashboard, analytics, CSV/PDF export, ledger tracking, notification alerts, and AI-assisted document or transaction risk explanation are included in the project. This project demonstrates secure digital trade finance management through authentication, document verification, transaction monitoring, risk analysis, and reporting.

Keywords— Trade Finance, FastAPI, React, Risk Scoring, SHA-256, JWT Authentication, Role-Based Access Control, Analytics Dashboard.

I. INTRODUCTION

Trade finance activities require secure handling of documents, transactions, users, approvals, alerts, and risk information. Manual trade finance processing can become difficult because documents need verification, transaction status must be tracked, and different users such as banks, corporates, auditors, administrators, and normal users require different access permissions.

To address these needs, this project introduces “Secure Trade Finance Management System.” The system provides a digital platform for managing trade finance documents and transactions through a web interface. It supports user registration, login, role-based access, document upload, hash verification, transaction creation, transaction status updates, alerts, ledger entries, risk score calculation, analytics dashboards, compliance reports, and export features. The project combines frontend development, backend APIs, database models, security logic, risk analysis, and reporting to support secure trade finance operations.

II. LITERATURE REVIEW

Trade finance management has evolved into a secure digital system that uses transaction management, document verification, blockchain ledger tracking, risk analysis, alerts, and audit monitoring to improve transparency, security, and efficiency in trade finance operations.

A. Traditional Trade Finance Management

Earlier Traditional trade finance management depends on manual document checking, separate records, and direct communication between parties. These methods may be slow and difficult to track. The project improves this process by storing users, documents, transactions, ledgers, alerts, and reports in a structured digital system.

B. Digital Document Integrity

Document verification is important in trade finance because documents such as invoices, letters of credit, bills of lading, purchase orders, certificates of origin, and insurance certificates must remain unchanged during the transaction process. In this project, document integrity is maintained using SHA-256 hashing. When a document is uploaded, the system generates a unique hash value from the file content and stores it with the document record. Later, the stored hash can be compared with an expected hash value to verify whether the document has been modified or not. The system also maintains ledger entries for document actions such as issued, amended, and verified, which helps track document activity and improves transparency between users.

C. Data Analysis and Visualization for Visa Trends

Risk analysis helps banks, administrators, and auditors review transactions and users based on possible risk factors. In this project, the system calculates risk scores using internal signals such as transaction count, total transaction volume, disputed transactions, dispute ratio, and unresolved alerts. The final risk score is stored with a rationale so users can understand the reason behind the score. The system also provides analytics dashboards, trade volume reports, disputed transaction counts, active user counts, average risk scores, CSV export, and PDF export, which support monitoring, reporting, and decision-making in trade finance management.

III. PROPOSED SYSTEM ARCHITECTURE & FEATURES

The Secure Trade Finance Management System is organized into multiple layers.

A. User Interface Layer

The user interface layer is developed using React and provides the main interaction point for users. It includes pages such as home, login, signup, dashboard, alerts, risk dashboard, ledgers, profile, document upload, documents, transactions, transaction creation, transaction details, integrity dashboard, compliance reports, settings, and users. The interface allows users to access trade finance functions in an organized way. Protected routes are used so that only authorized users can open specific pages based on their role.



B. Authentication and Role Management Layer

The authentication and role management layer controls user access to the system. The backend provides APIs for user registration, login, and token refresh. Passwords are securely stored using hashing, and JWT access and refresh tokens are used to authenticate users after login. The system supports Admin, Bank, Corporate, Auditor, and User roles. Each role has different permissions, and role checks are applied to restrict unauthorized actions.

C. Document Management Layer

The document management layer handles trade finance document operations. Users can upload documents by selecting document type, document number, and optional expiry date. The backend generates a SHA-256 hash for each uploaded file and stores the document record. Documents can be listed, viewed, updated, deleted, downloaded, analyzed, and verified using hash comparison. Ledger entries are also created for document actions, which helps maintain a clear record of document activity.

D. Transaction Workflow Layer

The transaction workflow layer manages trade transactions between buyer and seller users. A transaction can include buyer, seller, linked document, amount, currency, description, status, and decline reason. Transactions are created with an initial pending status and can move through valid status changes such as in progress, completed, disputed, or declined. Every status change is stored in transaction history. Alerts are generated for transaction creation, updates, disputes, and declines so related users can stay informed.

E. Risk Scoring and AI Assistance Layer

The risk scoring and AI assistance layer helps evaluate transaction and user risk. Risk scoring uses internal signals such as transaction records, total transaction volume, disputed transaction count, dispute ratio, and unresolved alerts. It also uses external signal values based on country and industry input. The final risk score is stored along with rationale and signal details. The project also includes AI-assisted document analysis and transaction risk question answering. If Gemini API access is not available, the system provides local fallback output.

IV. METHODOLOGY AND ALGORITHMIC DESIGN

The Secure Trade Finance Management System is based on secure document handling, transaction workflow management, risk scoring, alerts, and analytics. The system takes inputs such as user details, document data, buyer and seller information, amount, currency, country code, and industry. These inputs are processed to verify documents, manage transactions, calculate risk scores, generate alerts, and display dashboard reports.

A. Document Hashing

The Document hashing is used to protect uploaded trade finance documents. When a file is uploaded, the system generates a SHA-256 hash from the file bytes and stores it with the document record. During verification, the stored hash is compared with the expected hash value to check whether the document is unchanged.

$Document\ Hash = SHA256(File\ Bytes)$

B. Transaction Status Flow

The transaction status flow manages trade transaction progress. A transaction starts with Pending status and can move to In Progress, Disputed, Declined, or Completed based on valid rules. Each status change is saved in transaction history, and alerts are generated for related users.

C. Risk Score Calculation

Risk score calculation evaluates user or transaction risk using internal and external signals. Internal signals include total transactions, transaction volume, disputed transactions, dispute ratio, and unresolved alerts. External signals include trade barrier, financial stress, and logistics risk factors based on country code and industry. The final risk score is calculated using 65% internal score and 35% external score.

$Final\ Risk\ Score = Internal\ Score \times 0.65 + External\ Score \times 0.35$

D. Algorithm

Visa Processing Time Prediction Pipeline

Step 1: User Authentication

The process starts with user authentication where the user registers or logs in to the system. During registration, user details such as name, email, password, and role are stored securely in the database, and the password is encrypted using hashing techniques to ensure safety. When the user logs in, the backend verifies the credentials, and if they are correct, it generates an access token and a refresh token (usually JWT-based).

Step 2: Role Verification

After authentication, the system checks the role assigned to the user before allowing access to specific features. Each user is assigned a role such as admin, bank officer, auditor, or trader, and based on this role, permissions are controlled using role-based access control (RBAC). This ensures that only authorized users can perform certain actions, such as admins managing system-level operations or traders creating transactions, while restricting unauthorized access.

Step 3: Document Upload

In this step, the user uploads trade-related documents such as invoices, bills of lading, or certificates along with important metadata like document type, document number, and expiry date. The system stores the uploaded file securely in local or cloud storage and generates a unique hash (such as SHA-256) for the document. This hash acts as a digital fingerprint to ensure the document has not been tampered with, as even a small change will alter the hash value.

Step 4: Ledger Entry Creation

After a document is uploaded or an important action is performed, the system creates a ledger entry to maintain a permanent record. This entry includes details such as user ID, action type, timestamp, related document or transaction ID, and document hash. The ledger ensures transparency and provides a complete audit trail of all system activities, which is essential for tracking and compliance.

Step 5: Transaction Creation

The user then creates a transaction by entering details such as buyer, seller, amount, currency, optional document references, and a description of the trade. The system stores this transaction in the database with a unique transaction ID and assigns an initial status such as "Pending." This transaction becomes the central entity around which verification, tracking, and processing are performed.

Step 6: Status Update and Alert Generation

Once the prediction is generated, the system saves the new application details into the visa dataset. It also records the prediction information in a separate prediction history file. This stored data allows users to view previous predictions and helps maintain updated historical records. The saved information can also support future analysis and improvement of the prediction system.

Step 7: Risk Score Calculation

In this step, authorized roles such as admins, auditors, or bank officers calculate a risk score for transactions or users. The system evaluates multiple internal factors like user history, transaction behavior, and document validity, along with external factors such as fraud databases and compliance signals. Based on these inputs, a risk score is generated and categorized as low, medium, or high risk to help identify potentially suspicious activities.

Step 8: Risk Analytics and Export

The system dashboard shows key analytics like transactions, user activity, risk, and trade trends, and supports exporting reports in CSV or PDF format.

V. EXPERIMENTAL EVALUATION AND RESULTS

A. Experimental Setup

The project was evaluated using its implemented backend APIs, frontend pages, database models, and reporting modules. The evaluation focused on authentication, document handling, transaction workflow, alerts, risk scoring, integrity checks, dashboard metrics, and export features.

B. Quantitative Evaluation Metrics

- **Authentication Check:** Login, token creation, refresh token, and protected routes were checked.
- **Document Integrity Check:** Uploaded document hash values were compared with expected hash values.
- **Transaction Workflow Check:** Transaction creation, status update, history storage, and alerts were checked.
- **Risk Score Check:** Internal and external signals were used to generate final risk scores.
- **Reporting Check:** Dashboard metrics, CSV export, and PDF export were checked. data.

C. Discussion

The system supports secure trade finance management through structured user roles, protected APIs, document hashing, transaction history, alerts, ledgers, risk scoring, and analytics. The dashboard and reporting modules help users monitor transactions, risk values, disputes, and trade volume.

VI. CONCLUSION AND FUTURE WORK

This project presented Secure Trade Finance Management System, a web-based platform for handling trade finance documents, transactions, users, alerts, ledgers, risk scores, and reports. The system uses React, FastAPI, PostgreSQL, Redis, JWT authentication, SHA-256 hashing, and role-based access control to support secure and organized trade finance operations.

Future work can continue improving the existing project areas such as document verification, risk scoring, AI-assisted analysis, analytics reporting, caching, and role-based workflow management.

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